EXPLANATORY REPORT

Measurement-Free Memory Cost

Noisy records, quantum lifetime, and continuing non-Clifford processing

The question. Suppose every qubit that stores data or a controller record is noisy. Clean replacement states are available, and ideal Clifford operations may be performed between noise rounds. How much additional processing is needed to preserve an unknown quantum state?

With Clifford operations and resets alone, the best possible lifetime grows only logarithmically with physical width. A sufficiently large archive of noisy syndrome records attains this order. At more economical width, preserving high fidelity for long times requires continuing non-Clifford processing. The necessary gate count grows with both the logical payload and the reliability demanded over the storage horizon. At fixed positive noise, even a one-qubit payload requires order $D\ln(D/\epsilon)$ continuing processing when width is $O(\ln(D/\epsilon))$.

> **Fresh states can remove entropy. They do not, by themselves, supply the coherent processing needed to decide and apply corrections inside this memory.**

The lower bound is complemented by three storage constructions. A noisy archive defers nonlinear decoding to the final endpoint and uses exponential width. A concatenated five-qubit construction corrects during storage with at most 42 T or T-dagger gates per node and a polylogarithmic reliability overhead. A coherently compiled, constant-iteration Tanner-code decoder removes that overhead for $0 < \gamma \leq 1/20$ after a fixed number of inner five-qubit levels: its total noisy width is $O(k + \ln(J_+/\epsilon))$ and its prescribed T count is $O(nJ)$ for $J = O(\gamma D + 1)$ and $J_+ = \max(1, J)$. A rank-extinction converse makes this processing order necessary for every payload at fixed positive noise and width proportional to $k + \ln(D/\epsilon)$, once storage is sufficiently long. Numerical optimality, uniform weak-noise sharpness and practical constants remain open.

Both the converse and the constructions permit ideal global operations between noisy intervals. Storage duration counts noise rounds, separately from elementary gate depth.

Task	Preserve arbitrary unknown quantum information and its reference correlations.
Resources	Noisy physical width, number of noise rounds, and internal T/T-dagger count.
Central mechanism	Noise contracts a binary envelope of backward Pauli observables; non-Clifford gates can replenish it only at a bounded rate.

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Reading route. Sections 1–3 define the task and the forgetting mechanism. Sections 4–6 turn that mechanism into quantitative payload and reliability budgets. Section 7 explains the noisy archive; Section 8 gives both the five-qubit and Tanner-code coherent constructions. Sections 9–10 compare their resources and state the remaining questions. Section 11 derives a delayed-basis guessing premise, retains exponential rank moments, and obtains a finite and asymptotic decay guarantee. The appendix records the finite checks and a compact notation guide.

Conventions. All systems are finite-dimensional. A quantum channel is a completely positive, trace-preserving linear map, abbreviated CPTP. Logarithms written as ln are natural; $\log_2$ has base two. Probabilities, gate counts, qubit counts, and numbers of rounds are dimensionless. No physical clock time is assigned to a round.

1 What the memory must preserve

1.1 Why a classical record is part of the device

A classical bit can be copied, compared, and restored by majority voting. An unknown quantum state cannot be copied. Quantum error correction instead extracts a *syndrome*: information about the error that does not reveal the logical amplitudes. A decoder converts that syndrome into a correction.

In an ordinary measured architecture, a protected classical computer may retain syndrome histories and evaluate the correction rule. Here every retained record occupies noisy physical qubits in the

memory. Nonlinear processing during storage must be implemented by allowed gates on those registers. Storing the record in the computational basis does not protect it from noise.

The input may be entangled with an external reference, and recovery must retain that entanglement. Preserving a few expectation values or a measured classical description does not solve this task.

1.2 Width, rounds, and allowed operations

Let n be the maximum number of physical qubits present during storage, $k \geq 1$ the number of logical qubits, and $D \geq 0$ the number of noisy rounds. Width includes data, syndrome records, temporary workspace, and controller registers. An inactive slot may be reset before reuse. A preparation device or classical controller that retains information between rounds also belongs to the width.

Each round independently applies the product channel $\mathcal{N}_\gamma^{\otimes n}$, where

$$\mathcal{N}_\gamma(X) = (1-\gamma)X + \gamma \frac{I}{2} \operatorname{Tr} X, \qquad 0 < \gamma < 1. \tag{1.1}$$

The argument X in this formula is any single-qubit operator; for a density matrix it has trace one. Set

$$q = 1 - \gamma, \qquad \kappa = -\ln q > 0. \tag{1.2}$$

The parameter γ is the probability of *complete depolarization* in this mixture. Equivalently, the Pauli probabilities are $1 - 3\gamma/4$ for identity and $\gamma/4$ each for X, Y, Z. The resulting bit-flip probability for a computational-basis record is $\gamma/2$; it must be distinguished from the depolarization probability.

Between rounds, the fixed circuit may apply arbitrary ideal global Clifford unitaries, replacement resets, and counted T or T-dagger gates. A Clifford unitary maps a Pauli operator to another Pauli under conjugation, up to phase. A T gate is $\operatorname{diag}(1, e^{i\pi/4})$; its inverse is T-dagger. The total number of these internal gates is m_T.

A reset on subsystem A is exactly

$$\mathcal{R}_{\tau,A}(X) = \tau_A \otimes \operatorname{Tr}_A X, \tag{1.3}$$

with the original tensor ordering restored. The replacement state τ_A is independent of the input and of the retained subsystem. It may be mixed, nonstabilizer, or entangled within A. Correlated preparations at different times cannot provide an uncounted memory for the unknown input.

There is no postselection, free external feedback, or data-dependent external schedule. Coherent control is permitted when its registers and gates are counted. Processing between successive noise layers is ideal, so its elementary gate depth does not automatically produce additional noisy rounds.

1.3 The charged interval and the fidelity criterion

An arbitrary initial encoder $\mathcal{E}$ and arbitrary final decoder $\mathcal{D}$ lie outside the charged storage interval. With storage channel Φ_D, the recovered logical channel is

$$\Lambda = \mathcal{D} \circ \Phi_D \circ \mathcal{E}. \tag{1.4}$$

There is no protected quantum bypass between the endpoints. An endpoint may do extensive computation after storage ends; it supplies no protected memory during the preceding rounds.

Let $K = 2^k$ and define a normalized maximally entangled state between the logical input and a reference:

$$|\Omega_K\rangle = \frac{1}{\sqrt{K}} \sum_{j=1}^{K} |j\rangle|j\rangle. \tag{1.5}$$

The lower bounds assume Bell entanglement fidelity

$$F_{\mathrm{e}}(\Lambda) = \langle \Omega_K | (\mathrm{id} \otimes \Lambda)(|\Omega_K\rangle\langle\Omega_K|) | \Omega_K \rangle \geq 1 - \epsilon. \tag{1.6}$$

This convention uses overlap with the pure target, without taking a square root. The constructions prove the stronger worst-case bound

$$\delta_\diamond(\Lambda) := \frac{1}{2} \left\| \Lambda - \mathrm{id}_K \right\|_\diamond \leq \epsilon. \tag{1.7}$$

The diamond norm allows an arbitrary reference system: it bounds the trace-norm error of the output even when the input is entangled. Trace distance is half the trace norm, with $\|X\|_1 = \mathrm{Tr}\sqrt{X^\dagger X}$. Equation (1.7) implies (1.6), because a projector onto the target state is a valid measurement effect.

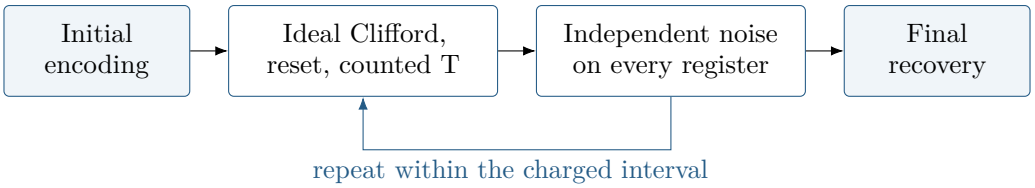


Figure 1: Every register carrying information between storage rounds is noisy. The repeated interval alternates ideal processing with independent noise on all registers; initial encoding and final recovery lie outside the charged interval.

2 The observable description of memory

2.1 Pauli labels are binary vectors

The single-qubit Pauli matrices I, X, Y, Z form an operator basis. On n qubits, their tensor products form a basis with 4^n elements. Ignore global phases and label a Pauli by

$$v = (x, z) \in W_n := \mathbb{F}_2^{2n}, \tag{2.1}$$

where (x_j, z_j) equals $(0,0), (1,0), (0,1), (1,1)$ for I, X, Z, Y, respectively. Adding labels modulo two corresponds to multiplying Paulis up to phase. Their commutation is described by

$$[v, w] = x \cdot z' + z \cdot x' \pmod 2, \qquad w = (x', z'). \tag{2.2}$$

A value zero means commuting, and one means anticommuting.

A Clifford acts as an invertible binary linear map preserving this form. It may spread a Pauli over many qubits while retaining a single Pauli label. Backward propagation can therefore bound the number of independent labels through which the input can influence the output.

2.2 Backward propagation tests input dependence

For a channel Φ, its adjoint satisfies

$$\mathrm{Tr}[O\Phi(\rho)] = \mathrm{Tr}[\Phi^\dagger(O)\rho]. \tag{2.3}$$

Thus $\Phi^\dagger(O)$ identifies which input observables can affect a chosen output measurement. If every output observable pulls back to a scalar multiple of identity, the output is independent of the input. The channel is then a replacement channel $X \mapsto \tau\,\mathrm{Tr}\,X$.

The backward calculation propagates all output observables *together*. For an operator O, its Pauli support is the set of labels with nonzero coefficients in its Pauli expansion. An envelope $V \leq W_n$ is a binary linear subspace containing every support encountered at a cut. Its dimension

$$r = \dim_{\mathbb{F}_2} V \tag{2.4}$$

counts independent binary labels. It is a proof quantity, not a number of stored logical qubits.

The envelope may contain labels absent from the actual operator supports. A reset can make coefficients vanish and leave a support set that is no longer linear. Enclosing that set in a linear space preserves an upper bound on input dependence without assuming that the surviving support is itself a subspace.

For the full physical output, initialize $V = W_n$, of dimension $2n$. If a particular final decoder is included, a smaller terminal envelope can sometimes be used. We write its dimension as $M \leq 2n$, and use $M = 2n$ unless the decoder gives a proved improvement.

2.3 How free operations update the envelope

Clifford conjugation maps V invertibly, preserving its dimension. A reset deletes coordinates of the reset subsystem. Indeed,

$$\mathcal{R}^{\dagger}_{\tau,A}(P_A \otimes P_B) = \mathrm{Tr}(\tau_A P_A) I_A \otimes P_B. \tag{2.5}$$

Every surviving label is the projection of the original label onto the retained coordinates, even when some scalar coefficients vanish. This projection is linear and cannot increase dimension. An arbitrary joint τ_A, including a magic state, changes the coefficients without adding independent binary labels under the reset.

Complete depolarization on a set E of qubits kills every Pauli with a nonidentity factor on E. Let π_E retain the two coordinates of the qubits in E. The backward envelope becomes

$$V \longmapsto V \cap \ker \pi_E. \tag{2.6}$$

An unrecorded Pauli measurement similarly removes anticommuting labels. This extension does not supply an external measurement record or a free adaptive decoder.

3 Why Clifford processing eventually forgets

3.1 Noise as an unobserved random channel

Equation (1.1) is a mixture of identity with probability q and complete depolarization with probability γ. Sampling these choices gives a CPTP circuit whose average is the original noisy channel. The sampled choices are used only in the proof; the memory and decoder receive no corresponding flags.

At one backward noise layer, the set E contains each qubit independently with probability γ. The expected surviving rank of any fixed r-dimensional subspace V obeys

$$\mathbb{E}_E \dim(V \cap \ker \pi_E) \leq qr. \tag{3.1}$$

Choose a full-row-rank $r \times 2n$ binary generator matrix G for V, and select r independent columns. Restrict G to the columns belonging to E, obtaining G_E. Every selected column whose qubit lies in E remains part of an independent subset of G_E. If Z_E counts those selected columns, then

$$\mathrm{rank}\, G_E \geq Z_E, \qquad \mathbb{E} Z_E = \gamma r. \tag{3.2}$$

The two coordinates of a qubit are selected together by the noise, but linearity of expectation does not require their indicators to be independent. Rank–nullity gives

$$\dim(V \cap \ker \pi_E) = r - \operatorname{rank} G_E, \tag{3.3}$$

which proves (3.1).

The envelope arriving at a backward noise layer depends only on noise later in forward time. It is therefore independent of the current layer. Conditioning on that envelope lets us iterate the contraction. Clifford operations and resets between layers cannot increase rank.

3.2 A replacement component gives a reference-stable bound

For the full storage channel, the terminal envelope has $M = 2n$. A smaller decoder-specific M applies to storage followed by that decoder. In this subsection, Φ_D includes the fixed decoder when that smaller envelope is used; the bound then concerns the decoded storage channel.

With no internal T gates, the final backward rank r_b in sample b satisfies

$$\mathbb{E}_b r_b \le M q^D. \tag{3.4}$$

Because a positive integer rank is at least one,

$$p := \Pr(r_b > 0) \le \min\{1, M q^D\}. \tag{3.5}$$

Every branch with $r_b = 0$ is exactly a replacement channel. Grouping branches yields

$$\Phi_D = (1-p)\mathcal{R}_{\tau_G} + p\Psi, \tag{3.6}$$

where Ψ is CPTP and τ_G is the average replacement state on the zero-rank event. Terms of zero weight can simply be omitted.

Fix any input state ω_0 and set $\tau = \Phi_D(\omega_0)$. Then

$$\Phi_D - \mathcal{R}_\tau = p(\Psi - \mathcal{R}_{\Psi(\omega_0)}). \tag{3.7}$$

The diamond distance between any two channels is at most two. Consequently,

$$\boxed{\frac{1}{2}\|\Phi_D - \mathcal{R}_\tau\|_\diamond \le \min\{1, M q^D\}.} \tag{3.8}$$

This bound has no reference-dimension prefactor. Arbitrary encoding before storage and decoding afterwards cannot increase diamond distance.

If $M = 0$, the decoded channel is already a replacement channel. For $M > 0$, a sufficient depth to make the right side at most $\delta \in (0,1)$ is

$$D \ge \frac{\ln(M/\delta)}{\kappa}. \tag{3.9}$$

Taking $M = 2n$ gives a lifetime upper bound of order $\gamma^{-1} \ln n$ at weak noise, since $\kappa = \gamma + O(\gamma^2)$ as $\gamma \to 0$.

For a recovered logical channel near a replacement, Bell fidelity is at most $4^{-k} + \delta$. A replacement has Bell fidelity exactly $K^{-2} = 4^{-k}$, whatever output state it prepares. Hence any fixed fidelity target $F_0 > 4^{-k}$ is incompatible with arbitrarily long zero-T storage at fixed width.

The Clifford-memory obstruction and the stochastic Pauli approach are established ingredients. The simultaneous-rank argument sharpens the forgetting scale to a single logarithm and makes the reference-stable statement direct. The next sections use the same rank to quantify continuing processing.

4 High fidelity requires an envelope at every temporal cut

4.1 From binary rank to an operator algebra

For a binary subspace V, let

$$\mathcal{A}_V = \operatorname{span}_{\mathbb{C}}\{P_v : v \in V\}. \tag{4.1}$$

This is a unital $*$-algebra: products add labels, and adjoints preserve their span. Its complex dimension is 2^r because distinct Paulis are linearly independent. Binary dimension r and operator-space dimension 2^r must be distinguished.

Suppose a tail channel Θ, including final decoding, has adjoint range inside $\mathcal{A}_V$. Then $\operatorname{rank}_{\mathbb{C}} \Theta \le 2^r$. Composing any CPTP prefix Π cannot increase this rank:

$$\operatorname{rank}_{\mathbb{C}}(\Theta \circ \Pi) \le 2^r. \tag{4.2}$$

Here rank refers to the linear map on operators, rather than to an output density matrix or the state representing the channel.

4.2 The Bell-fidelity bottleneck

Kraus operators A_α express the action of a channel as a sum of terms $A_\alpha X A_\alpha^\dagger$. For a logical CPTP channel Λ, the corresponding matrix on the operator space is $\sum_\alpha \overline{A_\alpha} \otimes A_\alpha$, where the bar denotes complex conjugation. Its trace gives

$$\operatorname{Tr}_{\sup} \Lambda = \sum_\alpha |\operatorname{Tr} A_\alpha|^2 = K^2 F_{\mathrm{e}}(\Lambda). \tag{4.3}$$

Every eigenvalue of a CPTP superoperator has modulus at most one. For example, its powers are trace-norm contractions on Hermitian operators and are uniformly bounded on all operators after decomposing into Hermitian real and imaginary parts. An eigenvalue of modulus greater than one would contradict this boundedness.

The number of nonzero eigenvalues, counted with algebraic multiplicity, is at most the matrix rank, including for a nondiagonalizable channel. Since the superoperator trace is nonnegative,

$$K^2 F_{\mathrm{e}}(\Lambda) \le \sum_{\lambda \ne 0} |\lambda| \le \operatorname{rank}_{\mathbb{C}} \Lambda. \tag{4.4}$$

Together with (4.2), this gives the branchwise bound

$$F_{\mathrm{e}}(\Theta \circ \Pi) \le \min\{1, 2^{r-2k}\}. \tag{4.5}$$

Now fix any primitive cut, immediately before or after a gate or noise layer. Sample the noise in the future tail and average the earlier noise into the prefix. Independence makes each resulting complete logical map CPTP. Its envelope rank is an integer r_b, and its Bell fidelity satisfies (4.5).

For every real $r \ge 0$,

$$\min\{1, 2^{r-2k}\} \le 4^{-k} + (1 - 4^{-k})\frac{r}{2k}. \tag{4.6}$$

On $[0, 2k]$, the convex exponential lies below the line connecting its endpoint values. Above $2k$, the line is at least one. Apply this inequality to each branch *before* averaging. Bell fidelity is linear in the channel, so the actual fidelity condition implies

$$R_t := \mathbb{E}_b r_{b,t} \ge a := 2k\,\frac{1 - \epsilon - 4^{-k}}{1 - 4^{-k}} > 0, \quad 0 \le \epsilon < 1 - 4^{-k}. \tag{4.7}$$

The same floor holds at every cut. The expectation R_t need not be an integer, and individual branches may fall below a. A large envelope is necessary for fidelity; it is not sufficient to transmit a quantum state.

For one logical qubit, $a = 2(1 - 4\epsilon/3)$. At $\epsilon = 0.01$, it is approximately 1.9733. At the replacement-fidelity threshold $\epsilon = 3/4$, the floor vanishes. This example illustrates the floor's fidelity dependence without asserting that any circuit attains it.

4.3 A complementary test for genuinely quantum information

The restricted symplectic form on V has even rank $2s$. Its radical consists of labels in V commuting with every label in V. If that subspace has dimension $c = r - 2s$, a Clifford change of coordinates puts the algebra in the form

$$\mathcal{A}_V \simeq M_{2^s} \otimes \mathrm{Diag}_{2^c} \otimes I_{2^{n-s-c}}. \tag{4.8}$$

Here M_d denotes all complex $d \times d$ matrices and Diag_d the diagonal matrices. The first factor can carry a coherent quantum subsystem. The second is classical, and the last carries no observable dependence in this algebra.

This structure has a direct channel interpretation. With symplectic orthogonal complement $V^\perp$, the map

$$\mathcal{E}_V(X) = \frac{1}{|V^\perp|} \sum_{w \in V^\perp} P_w X P_w \tag{4.9}$$

is CPTP and retains exactly the Pauli terms labelled by V. It is a self-adjoint projection onto $\mathcal{A}_V$. Thus an adjoint-range condition gives $\Theta = \Theta \circ \mathcal{E}_V$.

After this projection, any joint state with a reference is a mixture of classical sectors, each with at most a 2^s-dimensional coherent memory factor. A pure state's Schmidt rank is the number of nonzero singular values of its coefficient matrix across the memory–reference split. The projected state has Schmidt number at most 2^s: it admits a mixture of pure states with Schmidt rank bounded by 2^s. Local channels cannot increase that number.

Its overlap with $|\Omega_K\rangle$ is therefore at most $\min\{1, 2^{s-k}\}$. Indeed, a normalized pure state of Schmidt rank S has a coefficient matrix whose singular values have squared sum one. Their sum is at most $\sqrt{S}$ by Cauchy–Schwarz, so its squared Bell overlap is at most S/K.

In particular, a large commuting algebra has $s = 0$ and Bell overlap at most $1/K$. It may retain many classical labels while retaining no entanglement. This complements the stronger scalar floor obtained from total operator rank at small r.

5 The continuing processing lower bound

5.1 What one T gate can replenish

A fixed Pauli-axis rotation $U = e^{i\theta P_w}$ leaves commuting Paulis unchanged. An anticommuting P_v becomes a linear combination with labels v and $v + w$. A safe backward update is

$$V \longmapsto V + \langle w \rangle, \qquad \dim V' \le \dim V + 1. \tag{5.1}$$

A T or T-dagger gate is such a rotation about a single-qubit Z axis, up to global phase. It can add at most one binary generator. A Clifford or reset adds none.

More generally, assign a gate a proved rank-growth charge b_g if its update always satisfies $\dim V' \le \dim V + b_g$. Let $B = \sum_g b_g$. For actual Clifford+T circuits, take $B = m_T$. An arbitrary operation on a specified set of s_g qubits has a safe charge $2s_g$, because the outside labels are unchanged and

the inside coordinate space has that dimension. Such a structural charge is not automatically its T count.

5.2 A logarithmic balance law

Traverse the circuit backward, starting from rank M. Every cut has expected rank $R \geq a$. Noise gives $R' \leq qR$. A charged gate gives $R' \leq R + b_g$, so

$$\ln R' - \ln R \leq \ln(1 + b_g/R) \leq b_g/a. \tag{5.2}$$

Sum these logarithmic changes over all noise layers and gates. Free operations contribute nonpositive changes. Since the final backward rank is at least a,

$$\ln(a/M) \leq D \ln q + B/a. \tag{5.3}$$

It follows that

$$\boxed{B \geq a\,[D\kappa - \ln(M/a)]_+\,.} \tag{5.4}$$

Here $[x]_+ = \max\{x, 0\}$ and feasibility requires $M \geq a$. Taking $M = 2n$ gives the general decoder-independent result.

For T gates individually, their logarithmic increment is at most $\ln(1 + 1/a)$, giving the refinement

$$m_T \geq \frac{[D\kappa - \ln(M/a)]_+}{\ln(1 + 1/a)}. \tag{5.5}$$

Round up the right side when using an integer gate count. This refinement and the next section's finite-depth bound may both be imposed.

5.3 Interpreting the width allowance

At fixed error bounded away from the replacement threshold, $a = \Theta(k)$. For weak noise, $\kappa \sim \gamma$. Once $D\kappa$ exceeds the width allowance $\ln(2n/a)$ by a fixed factor greater than one, equation (5.4) requires

$$m_T = \Omega(k\gamma D). \tag{5.6}$$

The bound is extensive in logical payload and accumulated noise duration. The allowance grows as $\ln(n/k)$ for large width-to-payload ratios, but remains nonzero at finite rate when the error tolerance is nonzero. At finite encoding rate $k/n = \Theta(1)$, it corresponds to only $O(1/\gamma)$ rounds at fixed fidelity.

The conclusion does not charge the supply of magic states. Those states are explicitly allowed in replacement resets. It charges internal operations that expand the backward label envelope. Nor does it assert that the lower-bound number of gates is sufficient: an unsuitable code or recovery can fail despite a large gate count.

6 Finite depth, gate placement, and endpoints

6.1 An exact scalar relaxation

The placement of charged gates gives a sharper finite-depth bound. Number the noise layers in backward order by $j = 1, \ldots, D$. Let b_j be the total charge before backward noise layer j, and R_j the expected rank just after that layer. With $R_0 = M$,

$$R_j \leq q(R_{j-1} + b_j), \qquad R_j \geq a. \tag{6.1}$$

Charges after the last backward noise layer remain part of the physical total but cannot relax these earlier constraints.

Introduce a dominating scalar trajectory $x_0 = M$, $x_j = q(x_{j-1} + b_j)$. Its feasibility is equivalent to

$$\sum_{i=1}^{j} q^{j-i+1} b_i \geq a - Mq^j, \qquad b_i \geq 0, \quad j = 1, \ldots, D. \tag{6.2}$$

Minimizing $\sum_i b_i$ over real injections gives

$$\mathcal{L}(D, M, a, q) = \max\left\{0, \max_{1 \leq h \leq D}\left[\frac{a}{q} + (D-h)\frac{(1-q)a}{q} - Mq^{h-1}\right]\right\}, \tag{6.3}$$

$$B \geq \mathcal{L}(D, M, a, q). \tag{6.4}$$

Define $\mathcal{L}(0, M, a, q) = 0$.

The lower certificate follows from a weighted sum of the constraints. Fix an index h and multiply constraint j in (6.2) by zero for $j < h$, by $(1-q)/q$ for $h \leq j < D$, and by $1/q$ for $j = D$. The coefficient on b_i becomes q^{h-i} for $i < h$ and one for $i \geq h$, so it never exceeds one. The right side telescopes to the bracket in (6.3), certifying that lower bound for every h.

A matching feasible scalar trajectory waits until the reserve would fall below a, then replenishes only what is needed:

$$b_j = [a/q - x_{j-1}]_+ . \tag{6.5}$$

Write $t_0 = \lfloor \ln(M/a)/\kappa \rfloor$. If $D \leq t_0$, no injection is necessary in the relaxation. Otherwise the first positive injection is $a/q - Mq^{t_0}$ at layer $t_0 + 1$, followed by $(1-q)a/q$ per layer. Its total equals the certificate at $h = t_0 + 1$:

$$\mathcal{L} = \frac{a}{q} - Mq^{t_0} + (D - t_0 - 1)\frac{(1-q)a}{q}, \qquad D > t_0. \tag{6.6}$$

This also covers equality $Mq^{t_0} = a$: the next layer still needs replenishment.

> **The scalar optimization is exact for the necessary rank inequalities. It does not construct a physical memory or establish optimal T count.**

Actual ranks are random integers, gate support growth may be smaller than its charge, and a large envelope need not carry quantum information. These additional constraints can only increase the physical optimum.

The physical rank cap $2n$ gives another necessary condition when $D \geq 1$:

$$a \leq 2nq. \tag{6.7}$$

If this fails, the fidelity target is impossible regardless of gate count. In the feasible range $a \leq M \leq 2n$ and $a/q \leq 2n$, the scalar waiting-and-replenishing trajectory respects the cap. A convenient weaker consequence of (6.3) is

$$B \geq \left[\frac{(1-q)a}{q} D - (M - a)\right]_+ . \tag{6.8}$$

At fixed M, a, q, the slope of the scalar bound tends to $(1-q)a/q$. This is a statement about the bound; a separate erasure ceiling will prevent indefinite storage at fixed finite width.

6.2 Every interior interval has its own budget

Choose two primitive cuts enclosing a contiguous interval I. Let D_I count its full noise layers, B_I its internal charge, and M_I a deterministic upper bound on expected decoded-tail envelope rank at its later cut. Boundary gates are assigned according to the cuts so they are counted once. The same global fidelity guarantee supplies the floor a throughout the interval. Its noise is independent of the later suffix, so

$$B_I \geq \mathcal{L}(D_I, M_I, a, q), \qquad B_I \geq a\,[D_I\kappa - \ln(M_I/a)]_+ . \tag{6.9}$$

One can always use $M_I = 2n$. Gates in the later suffix can enlarge this envelope beyond the decoder's terminal value M, so a smaller terminal reserve cannot automatically be reused inside the interval. Thus gates placed entirely before or after an arbitrarily long interior segment cannot support that segment. The resource requirement concerns timing as well as a total count.

6.3 Why an arbitrary final decoder changes the reserve

A decoder with k output qubits need not have terminal support rank $2k$. Define its envelope by taking the binary span of every Pauli label in every pulled-back output Pauli. A single pulled-back observable can have many independent labels.

For example, a decoder measuring three qubits, computing majority, and preparing that output bit obeys

$$\mathcal{D}^\dagger(Z) = \tfrac{1}{2}(Z_1 + Z_2 + Z_3 - Z_1Z_2Z_3). \tag{6.10}$$

Its support spans three independent labels, though it outputs one qubit. The number of output observables is therefore not the dimension of their combined binary support.

For a Clifford decoder followed by discarding unused wires, the $2k$ output generators do pull back to $2k$ independent labels, so $M = 2k$ is valid. For arbitrary final decoding, $M = 2n$ is the safe choice. Moving nonlinear recovery to the free endpoint is permitted, but its support must be included in that endpoint envelope. Moving it inside the storage interval instead makes its gates part of the charge.

6.4 Reliability forces cumulative rank

A mean-rank floor captures the payload but does not by itself retain the probability that every nonconstant input direction is lost. That probability also constrains continuing work. Suppose $D \geq 1$ and $0 < \epsilon < 1 - 4^{-k}$. In this subsection put

$$\delta = \frac{\epsilon}{1 - 4^{-k}}, \qquad b = 1 - \delta, \qquad c_\gamma = \ln(1/\gamma) > 0. \tag{6.11}$$

Theorem 6.1 (Reliability-dependent processing)**.** *Every permitted memory with Bell fidelity at least* $1 - \epsilon$ *satisfies*

$$\boxed{m_T \geq \left[\gamma D b \max\left\{2k, \frac{[\ln(Db/\delta)]_+}{c_\gamma}\right\} - 2n\right]_+ .} \tag{6.12}$$

The bound is necessary; the finite-depth and individual-gate bounds remain complementary.

Keep extinction absorbing. Sample all independent replacement flags and propagate the envelope backward through the entire circuit. Use an update that preserves $V = \{0\}$. In particular, a rotation about Pauli axis w leaves V unchanged when every label in V commutes with w; otherwise its support is enclosed by $V + \langle w\rangle$. This still increases rank by at most one for a T or T-dagger gate, but never adds an axis to a zero envelope. Clifford operations, replacement resets and sampled

noise also preserve zero. Since every adjoint CPTP primitive maps identity to identity, this choice faithfully encloses the actual pulled-back observables.

Let r_j be the incoming rank immediately before the jth noise layer in backward order. It depends on the later suffix and is independent of the current layer's flags. For $r_j > 0$, choose r_j independent coordinate functionals whose restriction to V_j is injective, and collect the physical qubits containing these pivots. There are at most r_j such qubits. Replacing all of them removes every nonzero label of V_j: the survivor space is $V_j \cap W_{E^c} = \{0\}$. Hence the conditional probability of extinction at this layer is at least γ^{r_j}. This is a statement about the sampled channel decomposition, not a physical flag supplied to the memory.

Sum first extinction, rather than independent hazards. A fully sampled branch that reaches zero is input-independent and has Bell fidelity exactly 4^{-k}, including after arbitrary CPTP encoding and final decoding. Other branches have fidelity at most one. Linearity of fidelity therefore bounds the probability of eventual extinction by δ.

Write $S_j = \mathbf{1}_{\{r_j>0\}}$. Events in which the envelope first becomes zero at different noise layers are disjoint, because zero is absorbing. Free resets may also cause extinction; omitting those events only weakens the probability bound. Consequently

$$\sum_{j=1}^{D} \mathbb{E}[S_j \gamma^{r_j}] \le \delta. \tag{6.13}$$

Let $A = \sum_j \Pr(r_j > 0)$ and $B = \sum_j \mathbb{E} r_j$. A branch surviving the whole backward circuit is alive before every noise layer, so $A \ge Db$. Normalize the measure over the alive cut–branch pairs by A. Jensen's inequality for the convex function $e^{-c_\gamma r}$ then gives

$$\delta \ge A \exp(-c_\gamma B/A), \qquad B \ge \frac{A}{c_\gamma}[\ln(A/\delta)]_+ \ge \frac{Db}{c_\gamma}[\ln(Db/\delta)]_+. \tag{6.14}$$

The positive part uses $B \ge 0$. The last step is valid because $x[\ln(x/\delta)]_+$ is nondecreasing for $x \ge 0$.

Convert cumulative rank into charged gates. At noise layer j, expected rank drops by at least $\gamma \mathbb{E} r_j$. Every free operation has nonpositive rank increment, and each prescribed T/T-dagger slot adds at most one. Starting from rank at most $2n$ and ending at nonnegative rank yields

$$m_T \ge \gamma B - 2n. \tag{6.15}$$

The earlier fidelity floor also gives $\mathbb{E} r_j \ge 2kb$ at every incoming cut, so $B \ge 2kbD$. Combining this with (6.14) proves (6.12).

This reliability mechanism adapts established complete-erasure/Jensen reasoning for cumulative spacetime. Its role here is different: the hazard depends on backward Pauli rank, and the final balance charges T/T-dagger slots while permitting free Clifford operations and input-independent resets.

For any contiguous interval, the same proof uses its number D_I of noise layers and the same global extinction allowance δ. Retain the actual globally propagated suffix envelope at the interval's later cut, with expected rank at most $M_I \le 2n$; its suffix noise is independent of the noise inside the interval. Do not restart an unrelated envelope. Replace $D, 2n, m_T$ in the reliability part by $D_I, M_I, m_{T,I}$. This requires no claim that hazards at different times are independent.

6.5 Matching order for every payload at fixed noise

Fix $\gamma \in (0, 1/20]$, a width constant $C > 0$ and $0 < \epsilon_0 < 1/2$. Let $D \to \infty$, allow any $k \ge 1$ and $0 < \epsilon \le \epsilon_0$, and put $L = \ln(D/\epsilon)$. At width

$$n \le C(k + L), \tag{6.16}$$

the processing bound is

$$m_T = \Omega_{\gamma,C,\epsilon_0}(D(k+L)). \tag{6.17}$$

Indeed $\epsilon \le \delta \le 4\epsilon/3$ and $b \ge 1 - 4\epsilon_0/3 > 1/3$. Thus $\ln(Db/\delta) = L + O_{\epsilon_0}(1)$ uniformly over k, ϵ. For sufficiently large D, the maximum in (6.12) bounds a positive constant times $k + L$. Its positive $D(k + L)$ term then absorbs the initial reserve $2n \le 2C(k + L)$. The required duration threshold depends only on the fixed constants, rather than on k or ϵ.

The Tanner construction below attains $n = O(k + L)$ and $m_T = O_\gamma(D(k + L))$, so its processing order matches (6.17). For fixed k, this is order $D \ln(D/\epsilon)$; when k dominates the logarithm, it reduces to the payload order. The comparison fixes positive γ. The factor $1/\ln(1/\gamma)$ in the finite converse prevents interpreting it as an optimized uniform weak-noise coefficient.

Exact zero infidelity is impossible for finite width, $D \ge 1$ and positive noise: the first noise layer has a positive-probability complete-register replacement branch. At $D = 0$, the identity endpoint transmission needs no interior processing. These boundary cases are separate from the positive-error formula.

7 Zero-T storage by retaining a noisy syndrome history

7.1 What can be postponed

Errors in the noise model can also be sampled as independent Pauli labels. In each such history, a stabilizer code occupies a definite syndrome sector. Syndrome extraction can copy that sector label without copying the unknown logical state. If every error increment is correctable and the entire syndrome history can be recovered at the endpoint, physical feedback need not occur during storage.

Branching trees of copies preserve the noisy records until the endpoint, where majority voting and nonlinear decoding are allowed. During storage, the fixed schedule uses only Clifford operations and resets to $|0\rangle$.

7.2 A code correcting each noise increment

Assume $0 < \gamma \le 1/16$, $k, D \ge 1$, and $0 < \epsilon < 1$. In this section write the total memory width as N and the data-block width as n_0, to distinguish data from archive. Divide the rounds into batches of length at most

$$\tau = \left\lfloor \frac{1}{16\gamma} \right\rfloor, \quad J = \left\lceil \frac{D}{\tau} \right\rceil, \quad \Gamma = 1 - q^\tau \le \frac{1}{16}, \quad Q = 1 - \Gamma. \tag{7.1}$$

The final shorter batch has no larger noise. Each data qubit has a nonidentity Pauli increment with probability at most $p_* = 3\Gamma/4 \le 3/64$ per batch.

Stabilizer codes are common $+1$ eigenspaces of independent commuting Pauli generators. With s generators on n_0 qubits, the code space has dimension 2^{n_0-s} and encodes $k = n_0 - s$ qubits. Its distance is the minimum weight of a Pauli that commutes with every generator but acts nontrivially on the encoded state. Weight counts nonidentity tensor factors.

Choose distance at least $2w+1$, with $w = \lfloor n_0/16 \rfloor$, so every Pauli of weight at most w is correctable. Set

$$R_* = 1 - h_2(1/8) - \tfrac{1}{8}\log_2 3 > 0, \tag{7.2}$$

$$c_\Gamma = D_{\mathrm{bin}}(1/16 \| p_*), \tag{7.3}$$

$$n_0 = \left\lceil \max\left\{ \frac{k+1}{R_*}, \frac{\ln(2J/\epsilon)}{c_\Gamma} \right\} \right\rceil, \qquad s = n_0 - k. \tag{7.4}$$

Here $h_2(u) = -u\log_2 u - (1-u)\log_2(1-u)$ and

$$D_{\mathrm{bin}}(u\|v) = u\ln(u/v) + (1-u)\ln[(1-u)/(1-v)] \tag{7.5}$$

is binary relative entropy with natural logarithms. It is positive here because $p_* < 1/16$.

A finite counting argument supplies the code. Choose a uniformly random isotropic subspace $S \le \mathbb{F}_2^{2n_0}$ of dimension $n_0 - k$: every pair of its labels commutes. It describes s commuting stabilizers. Symplectic symmetry gives, for any nonzero label v,

$$\Pr(v \in S^\perp) = \frac{2^{n_0+k}-1}{2^{2n_0}-1} \le \frac{4}{3}\,2^{k-n_0}. \tag{7.6}$$

The number of Pauli labels of weight at most $2w \le n_0/8$ is at most $2^{n_0(1-R_*)}$. One way to see this is to set $r_* = 1/8$ and $y = r_*/[3(1-r_*)] < 1$ in the binomial generating function:

$$\sum_{j=0}^{2w} 3^j \binom{n_0}{j} \le y^{-r_* n_0}(1+3y)^{n_0} = 2^{n_0(1-R_*)}. \tag{7.7}$$

A union bound makes the probability of any nonzero such label in $S^\perp$ at most $(4/3)2^{k-R_* n_0} \le 2/3$. A code with the required distance therefore exists.

Let E_j denote the net data Pauli in batch j. Its weight has a binomial distribution with parameter at most p_*. A Chernoff bound and (7.4) give

$$\Pr\{\mathrm{wt}(E_j) > w\} \le e^{-n_0 c_\Gamma} \le \frac{\epsilon}{2J}. \tag{7.8}$$

Thus all data increments are within the guaranteed correcting radius except with total probability at most $\epsilon/2$. Code search and final minimum-weight decoding need not be efficient in this existence construction.

7.3 Extracting records without disturbing the payload

Let $g_1, \dots, g_s$ be the stabilizers, and let $\sigma(A)$ record the commutation bits of Pauli A with them. A Clifford encoder U maps input $|\psi\rangle|0\rangle^{\otimes s}$ to the code. To obtain b_{rep} replicas of the current syndrome, decode by $U^\dagger$, copy the s syndrome coordinates by CNOT into fresh zero targets, and reencode by U. In every Pauli branch,

$$A\mathcal{V}|\psi\rangle|0\rangle^{\otimes b_{\mathrm{rep}} s} \longmapsto A\mathcal{V}|\psi\rangle|\sigma(A)\rangle^{\otimes b_{\mathrm{rep}}}, \tag{7.9}$$

where $\mathcal{V}$ is the encoding isometry. The formula remains valid with an arbitrary reference attached to $|\psi\rangle$. The syndrome coordinates are definite computational-basis values conditional on the Pauli history; the logical amplitudes are not copied.

After extraction, records never interact with the data again. They only control CNOTs into fresh record children. Conditional on the data-error history, their dynamics is exactly a classical bit-flip process. Phase flips on disconnected computational-basis records do not affect the recovered bits.

Write $A_j = E_j \cdots E_1$, $\sigma_j = \sigma(A_j)$, and $\sigma_0 = 0$. At the start of batch $j \ge 2$, extract roots encoding σ_{j-1}. At that same boundary, each old live leaf creates three fresh children and is retired. The children and roots undergo the next batch's noise. There is no archive in batch 1. At the final endpoint, obtain σ_J directly from the data. The archive therefore stores precisely the intermediate syndromes $\sigma_1, \dots, \sigma_{J-1}$.

7.4 Why noisy trees retain a usable bias

For each extracted bit, start with b_{rep} independent replicas. Consider one tree, and let η_h be the probability that recursive ternary majority after h further generations misidentifies its *already noisy root*. With $e_* := \Gamma/2$ and $\eta_0 = 0$, a child differs from its parent after recursive reconstruction with probability at most $e_* + Q\eta_h$. Three independent children then give

$$\eta_{h+1} \le f_3(e_* + Q\eta_h), \qquad f_3(u) = 3u^2 - 2u^3. \tag{7.10}$$

If $\eta_h \le 1/4$, the argument is at most $(1+\Gamma)/4 \le 17/64$, and $f_3(17/64) < 1/4$. Induction proves $\eta_h \le 1/4$ for every age.

The extracted root has already suffered one batch of noise before producing descendants. Including that first noise, the probability of reconstructing the wrong *original extracted bit* is at most

$$e_* + Q/4 = \frac{1}{2} - \frac{Q}{4}. \tag{7.11}$$

The distinction matters because the target is the extracted syndrome, before its first noisy batch.

The b_{rep} replicas use disjoint trees and independent future noise after any slot reuse and reset. Conditional on the data history, their errors are independent. Majority over the replicas has failure probability at most

$$\exp(-b_{\mathrm{rep}}Q^2/8). \tag{7.12}$$

For $J \ge 2$, let $H_{\mathrm{arc}} = s(J-1)$ be the number of intermediate syndrome bits to archive. Choose b_{rep} as the smallest positive odd integer with

$$b_{\mathrm{rep}} \ge \frac{8}{Q^2} \ln \frac{2H_{\mathrm{arc}}}{\epsilon}. \tag{7.13}$$

There are H_{arc} archived bits, so all records are reconstructed correctly except with probability at most $\epsilon/2$. For $J = 1$, set $b_{\mathrm{rep}} = 0$ and omit the archive entirely.

7.5 Why the history corrects a quantum state

At the endpoint, reconstruct the intermediate syndromes and form successive differences

$$\widehat{\delta}_j = \widehat{\sigma}_j \oplus \widehat{\sigma}_{j-1}, \quad \widehat{\sigma}_0 = 0, \quad \widehat{\sigma}_J = \sigma_J. \tag{7.14}$$

If all archived bits are correct, syndrome linearity makes $\widehat{\delta}_j = \sigma(E_j)$. Choose a minimum-weight Pauli C_j with that syndrome. On the good data event, $\mathrm{wt}(C_j) \le w$ and $C_j^{-1}E_j$ has zero syndrome and weight at most $2w$, below the code distance. It is consequently a stabilizer up to phase. The product of these representatives cancels the accumulated data error on the logical subspace; Pauli reorderings introduce only global phases.

The final syndrome alone would not identify all correctable increments after many batches. The archive makes their *differences* available, which is why deferred feedback works here.

The joint good event has probability at least $1 - \epsilon$, independent of the logical state. In every good Pauli branch the recovered logical channel is exactly identity. Thus the full recovered channel is

$$\Lambda = (1-\beta)\,\mathrm{id} + \beta\Lambda_{\mathrm{bad}}, \qquad 0 \le \beta \le \epsilon, \tag{7.15}$$

with a CPTP bad-event channel. This proves $\delta_\diamond(\Lambda) \le \epsilon$ for arbitrary inputs and references, rather than only for classical codewords.

7.6 Counting the complete noisy workspace

Let L_j be the number of live archive leaves after batch j. Then

$$L_1 = 0, \qquad L_j = 3L_{j-1} + b_{\mathrm{rep}}s = \frac{b_{\mathrm{rep}}s}{2}(3^{j-1} - 1). \tag{7.16}$$

At a branching boundary, old parents and all fresh targets coexist. Counting only final leaves would miss this overlap. A fixed archive pool of $L_{J-1} + L_J$ slots suffices: reset targets outside the current live set, fan out, then retire the parents. Retired slots remain noisy and can later be reset and reused. The exact total width is

$$N = \begin{cases} n_0, & J = 1, \\ n_0 + b_{\mathrm{rep}}s(2 \cdot 3^{J-2} - 1), & J \geq 2. \end{cases} \tag{7.17}$$

All data, roots, leaves, parents, and reuse slots are included. Syndrome extraction and fanout compose to an ideal global Clifford at each batch boundary.

The constants in the coding and tree estimates can be chosen uniformly over $0 < \gamma \leq 1/16$. Since $J \leq 32\gamma D + 1$,

$$n_0 = O(k + \ln(J_+/\epsilon)), \tag{7.18}$$

$$b_{\mathrm{rep}} = O\left(\ln \frac{2J(k + \ln(J_+/\epsilon))}{\epsilon}\right), \tag{7.19}$$

$$N = O\left((k + \ln(J_+/\epsilon)) \ln \frac{2J(k + \ln(J_+/\epsilon))}{\epsilon}\ 3^J\right). \tag{7.20}$$

For fixed payload and error, $\ln N = O(1 + \gamma D)$. Inverting the construction at sufficiently large available width and combining it with (3.8) proves

$$\boxed{D_{\max}(N) = \Theta(\gamma^{-1} \ln N)} \tag{7.21}$$

at fixed k and fixed Bell-fidelity target $4^{-k} < F_0 < 1$, in the stated weak-noise range. This is sharp lifetime *order*. It does not optimize the width exponent, supply efficient code search, or remove the free final decoder.

8 Explicit coherent recovery with a charged gate budget

8.1 A five-qubit recovery node

The coherent construction performs nonlinear correction during storage instead of retaining an exponentially growing archive. It uses the five-qubit code with stabilizers

$$g_1 = XZZXI, \quad g_2 = IXZZX, \quad g_3 = XIXZZ, \quad g_4 = ZXIXZ, \tag{8.1}$$

and logical Paulis $\overline{X} = XXXXX$, $\overline{Z} = ZZZZZ$. These stabilizers commute and are independent; the two logical Paulis commute with the stabilizers and anticommute with each other.

A Clifford decoder maps the stabilizers to Z on four syndrome wires and the logical pair to X, Z on one remaining wire. Such a map exists by completing them to a symplectic basis. Under a Pauli error E, the syndrome wires are $s_1s_2s_3s_4$, where s_i is the commutation bit of E with g_i. The logical residual has bits $x(E) = [E, \overline{Z}]$ and $z(E) = [E, \overline{X}]$.

The fifteen one-site nonidentity Paulis have distinct nonzero syndromes. Their required logical correction bits are given exactly by the binary polynomials

$$f_X(s) = s_1 \oplus s_4 \oplus s_1s_3 \oplus s_2s_3 \oplus s_2s_4, \tag{8.2}$$

$$f_Z(s) = s_2 \oplus s_3 \oplus s_1s_2 \oplus s_1s_4 \oplus s_3s_4. \tag{8.3}$$

Products are binary AND and $\oplus$ is addition modulo two. Applying $X^{f_X(s)}Z^{f_Z(s)}$ to the decoded logical wire restores every error of weight at most one. The syndrome table in the appendix specifies the finite truth table behind these polynomials.

Each linear term is a Clifford controlled correction. The three quadratic terms of f_X use three Toffoli gates, each applying a NOT when both control qubits are set. Those of f_Z use three controlled-controlled-Z gates. A Hadamard on the target converts between these two three-qubit gates. Their seven-T implementation follows from the integer identity, for bits u, v, w,

$$\begin{aligned} 4uvw = u + v + w - (u \oplus v) - (u \oplus w) - (v \oplus w) \\ + (u \oplus v \oplus w). \end{aligned}$$

Applying phases $e^{i\pi/4}$ to the four positive parity terms and inverse phases to the three negative terms produces $e^{i\pi uvw}$, precisely a controlled-controlled-Z. Each parity can be computed by CNOTs into one of the original wires, phased, and uncomputed before processing the next term. This uses seven T/T-dagger gates and no additional qubit. Thus the recovery has the upper count

$$\boxed{6 \times 7 = 42 \quad \text{T/T-dagger gates per node.}} \tag{8.4}$$

This is a circuit upper bound, not a minimum. All controls are the original four syndrome wires, and the target is the logical wire. No additional work qubits are required. Reset the syndrome wires after correction; the order of the two conditional logical Paulis changes only a syndrome-dependent phase, irrelevant after those wires are discarded.

For every Pauli error, including uncorrectable ones, this circuit leaves a definite residual logical Pauli. That property is useful for concatenation.

8.2 Concatenation suppresses the residual error

Concatenate the code for $L \geq 0$ levels. One logical qubit occupies 5^L physical qubits. Correct the smallest blocks first, then their parents, up to the root, and reencode using the released and reset syndrome wires. The entire operation uses the original block width. The number of recovery nodes per logical block is

$$1 + 5 + \cdots + 5^{L-1} = \frac{5^L - 1}{4}. \tag{8.5}$$

Let p_ℓ bound the probability of a nonidentity residual at level ℓ. A parent succeeds whenever at most one of its five children has a residual error. Child error labels depend on disjoint independent physical noise samples, even for entangled logical inputs. A pair union bound gives

$$p_{\ell+1} \leq \binom{5}{2} p_\ell^2 = 10p_\ell^2, \qquad p_L \leq \frac{(10p_0)^{2^L}}{10}. \tag{8.6}$$

This deliberately conservative bound does not say that every weight-two error fails.

8.3 Batching makes the weak-noise dependence explicit

Assume integers $k, D \geq 1$, $0 < \epsilon < 1/2$, and $0 < \gamma \leq 1/20$. Choose

$$\tau = \left\lfloor \frac{1}{20\gamma} \right\rfloor, \qquad J = \left\lceil \frac{D}{\tau} \right\rceil \leq 40\gamma D + 1. \tag{8.7}$$

Perform recovery after every batch, including the last shorter one. A batch of $t \leq \tau$ rounds has effective strength $\gamma_t = 1 - q^t \leq t\gamma \leq 1/20$. Thus $p_0 = 3\gamma_t/4 \leq 3/80$ and

$$p_L \leq \frac{1}{10}(3/8)^{2^L}. \tag{8.8}$$

Batching gives a cost proportional to $\gamma D + 1$ rather than D, uniformly at weak noise, because recovery is performed after a batch instead of every arbitrarily short noise step.

For k logical blocks and J recovery batches, the probability of any residual logical error is at most kJp_L. Earlier logical Paulis do not change the probability of later, independently sampled errors. The event of no residual error is input-independent and acts as identity on all logical qubits and their reference. As in the archive construction, it gives an identity/bad-event channel mixture and hence $\delta_\diamond(\Lambda) \le kJp_L$.

8.4 Finite choices and the achieved resource law

With these parameters, set

$$A = \max\left\{1, \frac{\ln[kJ/(10\epsilon)]}{\ln(8/3)}\right\}, \qquad L = \lceil \log_2 A \rceil . \tag{8.9}$$

Then $kJp_L \le \epsilon$. The explicit construction satisfies

$$n = k5^L, \qquad m_T \le 42kJ\frac{5^L - 1}{4}, \qquad \delta_\diamond(\Lambda) \le \epsilon. \tag{8.10}$$

For $L = 0$ the recovery tree is empty and the gate count is zero; the same bound covers this case. Counting the last recovery makes the upper bound conservative and permits a final Clifford decoder.

Let $\alpha = \log_2 5$ and $\ell = 1 + \ln[k(\gamma D + 1)/\epsilon]$. Since $5^L \le 5A^\alpha$,

$$\boxed{n = O(k\ell^\alpha), \qquad m_T = O\big(k(\gamma D + 1)\ell^\alpha\big).} \tag{8.11}$$

The constants are uniform over this construction's stated weak-noise range. Its polylogarithmic overhead remains part of the five-qubit guarantee. The Tanner-code construction below removes that overhead in asymptotic order first within a separately proved, smaller direct noise range. A fixed number of inner five-qubit levels then extends the same resource order to $\gamma \le 1/20$.

For $k = 1$, $\gamma = 0.01$, $D = 1000$, and $\epsilon = 0.01$, the integer prescription gives $n = 125$ and $m_T \le 260{,}400$. It chooses $\tau = 5$, $J = 200$, $A \approx 7.75$, and $L = 3$, with conservative error certificate $20(3/8)^8 \approx 0.00782$. This example shows the cost of the explicit construction with its loose constants. It gives neither a practical optimum nor an estimate of hardware gate time.

8.5 A constant-iteration Tanner decoder

The next construction uses an inherited family of good quantum Tanner codes and its parallel decoder. Their code parameters and residual-error theorem are inputs; the new step is a fixed Boolean and coherent compilation with complete workspace and prescribed-gate accounting. It uses the same ideal processing model, with no retained syndrome history.

Write N for the Tanner *data-block length*, distinct from total noisy width n. Fix local length Δ, local relative distance d_r, product-expansion constant κ_c, and complementary local rates $\rho_c, 1 - \rho_c$ with $\rho_c \ne 1/2$. The inherited family has at least $R_0 N$ logical qubits and distance at least $d_0 N$, where

$$R_0 = (1 - 2\rho_c)^2 > 0, \qquad d_0 = \frac{d_r^2 \kappa_c^2}{256\Delta} > 0.$$

The subscript on κ_c distinguishes code expansion from the noise parameter $\kappa = -\ln q$.

Choose $0 < \delta_c < 1/18$. With

$$A_c = \frac{48}{\kappa_c \Delta}, \quad C_c = \frac{d_r^2 \delta_c^3 \kappa_c}{4096\Delta^2}, \quad g_c = \frac{1 - 18\delta_c}{16},$$

the inherited decoder, run for t parallel iterations, contracts each CSS error component modulo its stabilizer subgroup by

$$|e + f_t|_R \le a_t |e|_R, \qquad a_t = \frac{24}{5\kappa_c}(1 - g_c)^t, \quad A_c|e|_R \le \min(C_c N, d/\Delta).$$

Here the syndrome is perfect because it is freshly extracted during ideal processing; no syndrome survives from an earlier epoch. Fix t once so that $a := a_t \le 1/2$. This iteration count is independent of N, k, D, ϵ.

8.6 Every decoder decision belongs to a fixed circuit

A local view contains $b = \Delta^2$ bits. Fix lexicographic orders on local words and decompositions. Initial minimum-weight local corrections are Boolean functions of at most b current syndrome bits. During a parallel type substep, a candidate x is eligible exactly when

$$|Z| - |Z + x| \ge |x|/2 \quad \Longleftrightarrow \quad 4|Z \cap x| \ge 3|x|.$$

Selecting a maximum-weight eligible word, resolving ties, and minimizing its row/column decomposition defines one finite map from b bits to at most $2b$ bits. Enumerating its minterms implements all comparisons and minimizations. Each table uses at most $(b-1)2^b$ binary AND gates and at most $4b2^b + 4b$ total Boolean gates.

For both CSS components there are $(4 + 8t)N/b$ such tables and $O((1 + t)N)$ further XOR/copy operations. Four vertex types are processed in fixed order; views within each type are disjoint. Incidence connections are fixed wires. They require no data-dependent addressing, free lookup machine, or protected counter. This is a linear circuit-size argument at fixed b, t, rather than an appeal to classical RAM runtime.

Reset scratch to $|0\rangle$, extract the commuting stabilizer syndrome by Clifford gates, and evaluate the Boolean circuit reversibly into fresh work bits. Each AND is a Toffoli and each XOR/NOT is Clifford. Apply the computed correction masks with controlled Paulis, reverse the Boolean computation, and reset the original syndrome on a fixed schedule. A seven-T implementation of each Toffoli gives the prescribed bound

$$T_{\text{epoch}} \le 14(4 + 8t)\frac{(b-1)2^b}{b} N \le C_T N, \quad C_T = 112(1 + t)b2^b.$$

The factor two counts computation and uncomputation. It is charged even for an error-free syndrome. At most $4N$ syndrome bits and the linear Boolean workspace coexist with the data; a safe total is

$$n \le C_W N, \qquad C_W = 64(1 + t)b2^b.$$

Every slot remains noisy between epochs. Its noise is discarded by the next input-independent reset, not by protected storage. These constants can be enormous.

8.7 Why resetting the syndrome is logically safe

Let Π_s project onto syndrome s, and let $P_{f(s)}$ be the deterministic compiled correction. The coherent circuit followed by syndrome reset implements exactly

$$\mathcal{R}(\omega) = \sum_s P_{f(s)} \Pi_s \omega \Pi_s P_{f(s)}^\dagger.$$

This identity holds for arbitrary operators and after tensoring with an arbitrary reference. It is the intended measured syndrome-correction channel. On an encoded input subject to a fixed Pauli

trajectory, the syndrome is definite and contains no unknown logical amplitude. The retained original syndrome remains available while the Boolean circuit is uncomputed, even though correction changes the data's syndrome.

Put $\theta_c = \min(C_c, d_0/\Delta)$ and fix

$$u = \min\left\{\frac{\theta_c}{4A_c}, \frac{d_0}{8}, \frac{1}{8}\right\} > 0, \qquad v = \frac{au}{1-a} \le u.$$

If every fresh batch error has each CSS component of weight at most uN, and the previous residual components have reduced weight at most vN, then the incoming reduced weight is at most $(u+v)N \le 2uN$. The inherited theorem applies and leaves reduced weight at most $a(u+v)N = vN$. Induction starts at zero. Representatives of both components together have Pauli support at most $2vN \le d_0N/4 < d/2$, so one fixed endpoint decoder corrects every good trajectory. Minimum representatives are proof devices and are never computed by the controller.

8.8 Batching, available lengths, and the achieved order

The proved range is $0 < \gamma \le \gamma_* := u/2$. Take

$$h = \lfloor u/\gamma \rfloor \ge \frac{u}{2\gamma}, \qquad J = \lceil D/h \rceil \le 2\gamma D/u + 1.$$

A batch has retention q^h. Each X or Z component separately has probability $p_h = (1-q^h)/2 \le u/2$ at each data position. The components are correlated on the same qubit, so only separate binomial tail bounds are used. With $c = D_{\mathrm{bin}}(u\|u/2) > 0$, a union bound over both components and all batches gives

$$P_{\mathrm{bad}} \le 2Je^{-cN}.$$

No independence between the two residual components is assumed.

Suitable lengths have bounded gaps. For one fixed odd prime power Q and $\Delta = Q+1$, the underlying group construction provides every integer $i > 2$, with

$$N_i = \frac{\Delta^2}{2}Q^i(Q^{2i}-1), \qquad N_{i+1}/N_i < 2Q^3.$$

Thus one may round up to an available length satisfying

$$N \ge \max\{N_0, k/R_0, c^{-1}\ln(2J_+/\epsilon)\}, \quad J_+ = \max(1, J),$$

at a fixed multiplicative cost. The free encoder pads unused logical axes with fixed states. Every good input-independent Pauli trajectory is exactly corrected by the same endpoint decoder. Consequently $\Lambda = (1-P_{\mathrm{bad}})\,\mathrm{id} + P_{\mathrm{bad}}\Xi$ for a CPTP map Ξ, and

$$\boxed{n = O(k + \ln(J_+/\epsilon)), \quad m_T = O(nJ), \quad \delta_\diamond(\Lambda) \le \epsilon, \quad J = O(\gamma D + 1).} \tag{8.12}$$

For $D = 0$ no recovery is required. Constants depend only on the fixed code, decoder slack and iteration count, and length-gap factor, uniformly for $0 < \gamma \le \gamma_*$. This bare Tanner noise range can be far smaller than $1/20$. The following composition proves the extension; it is not inferred from an LDPC threshold.

When $k = \Omega(\ln(J_+/\epsilon))$ and $\gamma D \to \infty$, the construction has $n = O(k)$ and $m_T = O(k\gamma D)$. More generally, at fixed positive γ, its width and processing match the all-payload law (6.17), including the reliability-dominated case. Existing Tanner codes, parallel decoding, repeated-round correction, coherent syndrome correction and passive LDPC memory mechanisms remain inherited ingredients. The addition is their model-compatible compilation and resource consequence, with no claim of an optimal numerical coefficient or practical threshold.

8.9 Fixed-level inner protection extends the noise range

The outer decoder only needs a fixed tolerance, rather than an error shrinking with storage duration. Consequently a *fixed* number of inner five-qubit levels suffices to reach that tolerance throughout $0 < \gamma \le 1/20$.

For a batch of length

$$h = \left\lfloor \frac{1}{20\gamma} \right\rfloor, \qquad J = \lceil D/h \rceil \le 40\gamma D + 1,$$

the physical nonidentity Pauli probability is at most $3/80$. The elementary inner recursion $p_{\ell+1} \le 10p_\ell^2$ therefore gives

$$\eta_\ell = \frac{(3/8)^{2^\ell}}{10}, \qquad L_* = \min\{\ell \ge 0 : \eta_\ell \le u/2\}, \qquad W = 5^{L_*}.$$

The constants L_*, W depend only on the fixed outer family. They do not grow with k, D, ϵ or depend on γ in the stated range.

Encode each outer data qubit in W inner physical qubits. At every epoch, coherently decode and correct each inner block, apply the compiled outer recovery, and reencode by a Clifford circuit. Reset unused syndrome and workspace on the prescribed schedule. All of this processing lies in an ideal interval between noisy layers. Every persistent physical slot remains noisy and counted.

Disjoint inner blocks transform fresh independent physical Pauli noise into independent fresh Pauli errors across outer positions, each with total nonidentity probability at most $\eta_{L_*} \le u/2$. A pre-existing Pauli residual on the outer data acts as a logical Pauli of each inner block. It commutes with the inner stabilizers, leaves their syndrome unchanged, and passes through correction composed with the fresh residual label. The effective channel need not be isotropic, and its X/Z components need not be independent. Each component's error probability is at most $u/2$, which is exactly what the preceding marginal tail argument requires.

Thus the same outer invariant and $2Je^{-cN}$ tail certificate apply. A safe complete resource count is

$$n \le (W + C_W)N, \qquad m_T \le \left\lceil 42\frac{W-1}{4} + C_T \right\rceil NJ.$$

Inner correction contributes all $42(W-1)/4$ T slots per block per epoch; outer decoding contributes $C_T N$. Inner encoding and wire routing are Clifford. Outer scratch is reset before use and need not retain an encoded record between epochs. The available-length and padding argument is unchanged. Therefore (8.12) holds throughout $0 < \gamma \le 1/20$. This extends the proved range without restoring a reliability logarithm, at potentially enormous fixed cost.

9 Why more gates still require growing width

There is a limitation independent of the Clifford restriction. In the identity/depolarization decomposition, one noise layer completely depolarizes all n qubits with probability γ^n. Once that happens, all correlation with the original unknown input is gone. Input-independent subsequent operations and replacement states cannot restore it.

The probability that this never happens in D independent layers is $(1-\gamma^n)^D$. Every branch containing such a layer is a replacement channel from the logical input and has Bell fidelity 4^{-k}. Other branches have fidelity at most one. Thus, even with unrestricted internal gates,

$$\boxed{F_{\mathrm{e}} \le 4^{-k} + (1-4^{-k})(1-\gamma^n)^D.} \tag{9.1}$$

For $D \geq 1$ and $0 < \epsilon < 1 - 4^{-k}$, put $\eta = \epsilon/(1 - 4^{-k})$. High fidelity requires

$$(1 - \gamma^n)^D \geq 1 - \eta. \tag{9.2}$$

Since $1 - x \leq e^{-x}$, a useful necessary consequence is

$$\gamma^n \leq \frac{-\ln(1-\eta)}{D}, \quad n \geq \frac{\ln[D/(-\ln(1-\eta))]}{\ln(1/\gamma)} \tag{9.3}$$

whenever the numerator is positive. For fixed γ and small error this is logarithmic width in D/ϵ. It is an elementary necessary ceiling, not an attainable resource formula.

At any fixed finite width, the right side of (9.1) tends to 4^{-k} as $D \to \infty$. Consequently, the long-time slope of the processing lower bound must be interpreted along families with adequate growing width, not as a promise of indefinitely reliable storage in one finite device.

10 Comparing the storage resource guarantees

The constructions make different resource choices. The archive retains noisy syndrome history for nonlinear recovery at the endpoint; the five-qubit and Tanner constructions evaluate corrections during storage. Their guarantees and the necessary resource bounds are compared below.

Resource choice	Established result	Boundary of the claim
Clifford and resets only	Sharp lifetime order $\Theta(\gamma^{-1} \ln n)$ at fixed k, F_0	Weak noise; exponential archive width; unrestricted endpoint decoding
Five-qubit coherent correction	Explicit polylogarithmic reliability overhead for $\gamma \leq 1/20$	Modest local circuit; overhead remains in this construction
Tanner coherent correction	$n = O(k + \ln(J_+/\epsilon))$, $m_T = O(nJ)$, $J = O(\gamma D + 1)$	Range $\gamma \leq 1/20$ after fixed-level inner protection; fixed-noise order matching at width $O(k + \ln(D/\epsilon))$; enormous constants
Arbitrary internal gates	Full-register depolarization ceiling (9.1)	A necessary width condition, not a matching code construction

Table 1: The lower bounds, constructions, and ultimate width ceiling answer different parts of the same storage problem.

The common mechanism is the loss of independent binary directions in the output's dependence on the input. Clifford processing rearranges these directions and resets alter their coefficients, while a T gate can add at most one. High fidelity requires enough directions at every temporal cut, which yields both the total processing budget and its interval version.

The established ingredients include the qualitative memory obstruction, stabilizer and symplectic methods, the algebra and entanglement tools, random-code existence, tree broadcasting, five-qubit coding, concatenation, and exact Toffoli synthesis. The quantitative account developed here lies in the simultaneous envelope contraction, fidelity-dependent and finite-depth budgets, and the fully charged storage constructions. The self-contained derivations are not a claim that every ingredient is new.

Several questions remain. What is the optimum numerical dependence on width, fidelity, payload, and gate placement beyond the fixed-noise, width-restricted matching order? Can useful constants be proved for an explicitly instantiated decoder, or can the physical noise range be extended further? Can stronger algebraic information improve the numerical converse or close its remaining uniform weak-noise gap? A lower bound for one particular decoder would not answer these unrestricted questions.

The present guarantees do not include noisy elementary gates, restricted connectivity, finite processing time, correlated noise, or preparation faults. All constructions use the stated ideal processing model. A measured protected controller also changes the available resources and requires a separate analysis. These boundaries prevent a direct reinterpretation of D as hardware depth.

11 A delayed-basis score as an alternative premise

11.1 The device receives only a quantum challenge

The same rank ledger can be tested through a classical answer rather than a prescribed Bell-fidelity target. Choose independent uniform strings $X, \Theta \in \{0,1\}^k$ and supply the k-qubit state

$$|X\rangle_\Theta = \bigotimes_{i=1}^{k} H^{\Theta_i} |X_i\rangle, \qquad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Keep X secret and reveal only Θ after all charged storage. The final decoder may depend arbitrarily on Θ. Let $p = \Pr[\widehat{X} = X]$ be unconditional *whole-string* success, counting aborts as failures. Write $d = 2^k$ and $b_0 = 1/d$.

The width remains n and the number of independent depolarizing layers remains D. Every persistent controller is noisy, counted and processed through the allowed fixed primitives. Counting records does not make arbitrary feedback or data-dependent resets free. Input-independent resets and arbitrary endpoint decoding retain their earlier meaning, without protected side channels or postselection.

Start one common backward envelope at the full physical output *before* basis-dependent readout, with rank $M = 2n$. The guessed output length does not reduce this reserve to $2k$.

11.2 A quantum-plus-classical bottleneck at every cut

At a cut t, average earlier noise into a CPTP prefix Π_t. An independent future-noise realization b selects a CPTP tail $S_{b,t}$ to the full physical output. The Pauli algebra from Section 4 gives

$$S_{b,t} = S_{b,t}\mathcal{E}_V, \qquad \mathcal{A}_V \simeq M_{2^s} \otimes \mathrm{Diag}_{2^c} \otimes I_G, \qquad r = 2s + c.$$

The projection $\mathcal{E}_V$ factors as $J_V F_V$. The CPTP map F_V changes to canonical coordinates, dephases the c-bit factor, and discards G. The CPTP map J_V reinserts the maximally mixed gauge factor and reverses the coordinates. Thus $F_V \Pi_t$ is a genuine encoder into s quantum qubits and c classical bits. Composing J_V with the tail and final readout reproduces the actual outcome distribution for each basis.

Let $p_{b,t}$ be the actual endpoint decoder's success probability on the sampled channel. For this encoder, delayed-basis BB84 guessing obeys

$$p_{b,t} \le \min\{1,\, 2^{(s-k)/6},\, 2^{s+c-k}\}. \tag{11.1}$$

The first nontrivial bound permits unlimited classical records. The second uses the finite output dimension: at each fixed basis, d equiprobable output states satisfy $\rho_x \le I$, so their discrimination success is at most $\mathrm{Tr}\, I/d = 2^{s+c-k}$. This count uses the original normalized encoder, before any additional mathematical classical refinement of its states.

The noise flags are proof aids, not information supplied to the device. They select channels independently of the input and basis, so $p = \mathbb{E}_b p_{b,t}$ at every primitive cut. Each branch may be upper-bounded by its optimum decoder without physically revealing a flag.

For $k \geq 8$, the two bounds in (11.1) give

$$p_{b,t} \leq b_0 + (1 - b_0)\frac{r}{2k}, \qquad R_t \geq a_p := 2k\frac{[p - b_0]_+}{1 - b_0}. \tag{11.2}$$

One way to see the first inequality is to maximize the branch bound at fixed r. Relaxing s continuously gives 2^{r-k} up to $r = 5k/6$, then $2^{(r-2k)/7}$ up to $r = 2k$, and one thereafter. For $k \geq 8$, both convex exponential pieces lie below the chord from $(0, b_0)$ to $(2k, 1)$. Averaging that chord proves the floor. For every $k \geq 1$, a convenient weaker floor is $k[p - b_0]_+/(1 - b_0)$, obtained from the finite-output dimension bound alone. Smaller-block refinements can retain the full concave polygon instead.

11.3 Processing, timing, and the recovery comparison

Insert any positive score-derived floor into the earlier balance laws. In particular, (11.2) gives

$$B \geq a_p \left[D\kappa - \ln(2n/a_p)\right]_+, \qquad B \geq \mathcal{L}(D, 2n, a_p, q). \tag{11.3}$$

The individual-T refinement (5.5) also applies. Every contiguous interval has the same global floor, with its own D_I, B_I and entering reserve $M_I \leq 2n$. Gates outside the interval supply no additional reserve beyond that bound. For $k = 12, p = 0.4, n = 120, \gamma = 0.01, D = 2000$, the floor is $a_p \simeq 9.59648$ and the individual-gate refinement requires at least 171 internal T/T-dagger gates. Every 700-round contiguous interval requires at least 39 such gates by the same refinement; the scalar injection bound alone gives 37. These necessary conditions do not specify an achievable circuit.

The recovery comparison remains informative even at scores at or below one half, provided it is applied *before* noise averaging. For one complete branch channel, let F_{rec} be its best squared Bell-overlap fidelity, $F_{\mathrm{pg}} \leq F_{\mathrm{rec}}$ its pretty-good value, u its average pretty-good guessing success, and $p_* \geq 1/d$ its optimized guessing score. Define

$$g_d(p_*) = p_*^2 + \frac{(1 - p_*)^2}{d - 1}, \qquad A_\ell = \sum_{j=0}^{\ell} \binom{k}{j}(1 - 2^j\tau_\ell),$$

where $\tau_\ell = 2^{-\ell-1}$ for $\ell < k$ and $\tau_k = 0$. Established discrimination and recovery estimates give

$$\begin{aligned} g_d(p_*) \leq u, \qquad & F_{\mathrm{rec}} \geq p_*/d, \\ u \leq \tau_\ell + A_\ell F_{\mathrm{pg}}, \qquad & 0 \leq \ell \leq k. \end{aligned} \tag{11.4}$$

The underlying spectral reduction writes $u = \sum_v \lambda_v w_v$, where $w_v \geq 0$, $\sum_v w_v = 1$, and $w_v \leq F_{\mathrm{pg}}$. Its nonzero eigenvalues are 2^{-j} with multiplicities $2^j\binom{k}{j}$. Bounding all terms at or below τ_ℓ by that threshold, then applying the cap to the excess, gives $u \leq \tau_\ell + F_{\mathrm{pg}} \sum_{\lambda_v > \tau_\ell}(\lambda_v - \tau_\ell)$, which is the displayed estimate. At $\ell = 0$ it gives $u \leq (1 + F_{\mathrm{pg}})/2$. At $p_* = 1/2$, taking $\ell = 2$ gives a positive recovery-fidelity lower bound of order k^{-2}.

Each branch also has $F_{\mathrm{rec}} \leq \min\{1, 2^{r-2k}\}$. Combining these inequalities branchwise and then averaging reproduces (11.2) for every $k \geq 8$. Exact finite-rank checks also imply the weaker floor stated above for $1 \leq k \leq 7$. These mean-rank floors and their gate or interval consequences therefore already follow from the recovery route. The direct factorization provides a short operational proof; the comparison establishes neither an optimal attainable score nor dominance over every possible stronger direct bound.

11.4 A score also bounds extinction

For the full-output envelope, an extinct branch is input-independent before basis readout. Since the hidden string remains uniform and secret, such a branch has guessing success at most 2^{-k} (aborts can only lower it). For $2^{-k} < p < 1$, eventual extinction therefore has probability at most

$$\delta_p = \frac{1-p}{1-2^{-k}}, \qquad b_p = 1 - \delta_p. \tag{11.5}$$

The first-extinction and cumulative-rank argument in Section 6.4 then gives, for every challenge length $k \geq 1$,

$$m_T \geq \left\lceil \frac{\gamma D b_p}{\ln(1/\gamma)} [\ln(D b_p/\delta_p)]_+ - 2n \right\rceil_+ . \tag{11.6}$$

Here only the reliability part is substituted; no large-k mean-rank chord is assumed for a small challenge. For fixed k, fixed above-chance p and fixed positive noise, this forces order $D \ln D$ work at width $O(\ln D)$. The same inference can concern retained classical input dependence, so it does not alone certify coherent quantum storage. Neither the score nor this bound verifies the noise and controller premises.

11.5 Retaining the noise distribution through rank moments

The mean-rank chord discards information about which rank mixtures the product noise can generate. Retaining an exponential moment gives a stronger family of necessary processing budgets in the same model. For a fixed real $1 < z \leq 2$, define

$$M_z(t) := \mathbb{E}_b z^{r_{b,t}}, \qquad X_z(t) := \log_z M_z(t), \qquad \alpha_z := \frac{\ln(1-q+qz^2)}{2\ln z}. \tag{11.7}$$

Here $0 < \alpha_z < 1$. The common full-output envelope has $X_z = 2n$ at its terminal cut.

Choose r binary coordinate functionals that identify every vector in a rank-r envelope V. If $d_j \in \{0,1,2\}$ of these pivot coordinates belong to qubit j, then $\sum_j d_j = r$. For a set S of retained qubits, projection onto the pivots gives

$$\dim(V \cap W_S) \leq \sum_{j \in S} d_j.$$

Independent retention with probability q therefore implies

$$\mathbb{E}[z^{r'} \mid V] \leq \prod_j (1-q+qz^{d_j}) \leq (1-q+qz^2)^{r/2}. \tag{11.8}$$

The last inequality pairs one-pivot factors, using $(1-q+qz)^2 \leq 1-q+qz^2$; an unpaired factor is bounded by the square root. This argument uses the actual independent noise choices, with no assumption of rank concentration.

Conditioning on the later suffix makes its incoming envelope independent of the next backward noise layer. Averaging (11.8) and applying concavity of x^{α_z} yields

$$X'_z \leq \alpha_z X_z \quad \text{at noise}, \qquad X'_z \leq X_z + 1 \quad \text{at T or T-dagger}. \tag{11.9}$$

Clifford gates preserve rank, and an input-independent reset projects the support and cannot increase it. Both free primitives therefore satisfy $X'_z \leq X_z$. A gate of structural charge b_g gives the corresponding increment at most b_g. The circuit is still one fixed circuit across all noise branches; no mathematical flag is supplied to its controller.

11.6 A fixed recovery produces a floor for every moment

Apply the established guessing/recovery estimates to the *averaged* channel Φ from the challenge to the full physical output, before basis readout. For $p \geq 1/d$, put

$$F_0(p) := \max\left\{d^{-2}, \frac{p}{d}, \max_{0\leq\ell\leq k} \frac{[g_d(p) - \tau_\ell]_+}{A_\ell}\right\}. \tag{11.10}$$

For $p < 1/d$, use only $F_0 = d^{-2}$ and set the resulting floor to zero. There is a single CPTP recovery of Φ with squared Bell fidelity at least $F_0(p)$. Fix this recovery. At every primitive cut, the earlier noise is averaged into the prefix, while each sampled future tail factors through its Pauli algebra. Its recovered branch channel obeys $F_b \leq \min\{1, 2^{r_b-2k}\}$. Fidelity is linear in the recovered channel, so $\mathbb{E}_b F_b \geq F_0(p)$ at every cut. The recovery is common to the branches; it need not know the noise flags.

Set $b = 4^{-k}$ and $w = (F_0 - b)/(1 - b)$. The power $2^r = (z^r)^{\ln 2/\ln z}$ is convex in z^r for $z \leq 2$. Its endpoint secant gives

$$\min\{1, 2^{r-2k}\} \leq b + (1 - b)\frac{z^r - 1}{z^{2k} - 1},$$

also when $r \geq 2k$, since the right side is then at least one. Consequently every cut satisfies

$$X_z(t) \geq a_z^{\mathrm{rec}} := \log_z\Big[1 + w(z^{2k} - 1)\Big]. \tag{11.11}$$

At chance $F_0 = b$ this gives zero. Every above-chance score makes $F_0 \geq p/d > b$ and hence $a_z^{\mathrm{rec}} > 0$. For the fidelity task, a directly supplied squared Bell-fidelity guarantee may replace $F_0(p)$ in the same argument.

The direct quantum-plus-classical profile in (11.1) supplies a complementary floor throughout the above-chance range. Put

$$b_0 = 2^{-k}, \quad H_k = 2^{-k/6}, \quad t_0 = 5k/6, \quad Y_t = z^{t_0}, \quad Y_e = z^{2k}.$$

For $1 < z \leq 2^{1/7}$, both exponential pieces of its rank curve are convex functions of $Y = z^r$. Their endpoint chords are therefore upper bounds. Let $P_z(Y)$ be the concave polygon through

$$(1, b_0), \qquad (Y_t, H_k), \qquad (Y_e, 1),$$

continued constantly after Y_e; omit the middle point if it lies below the chord joining the other two. The branch-score bound and Jensen give $p \leq P_z(M_z)$. Its inverse is explicit. If the middle point is omitted,

$$L_z(p) = 1 + \frac{p - b_0}{1 - b_0}(Y_e - 1).$$

If it is retained, use

$$L_z(p) = \begin{cases} 1 + \dfrac{p - b_0}{H_k - b_0}(Y_t - 1), & b_0 < p \leq H_k, \\ Y_t + \dfrac{p - H_k}{1 - H_k}(Y_e - Y_t), & H_k \leq p \leq 1. \end{cases} \tag{11.12}$$

Thus $a_z^{\mathrm{dir}} = \log_z L_z(p)$ is an all-cut floor; set it to zero at or below chance. For each z, use the larger available direct or recovery floor and call it a_z. The earlier recovery comparison of mean-rank floors does not settle this comparison after the noise distribution is retained.

Let $\kappa_z = -\ln \alpha_z$. Equations (11.9)–(11.11) have exactly the form used in Sections 5–6. Thus, for each z with $a_z > 0$,

$$B \geq a_z[D\kappa_z - \ln(2n/a_z)]_+, \qquad B \geq \mathcal{L}(D, 2n, a_z, \alpha_z), \tag{11.13}$$

$$m_T \geq \left\lceil \frac{[D\kappa_z - \ln(2n/a_z)]_+}{\ln(1 + 1/a_z)} \right\rceil. \tag{11.14}$$

For T/T-dagger circuits, the second bound in (11.13) can also be rounded upward. Retain the largest applicable lower bound over z and the earlier mean-rank bounds. The necessary one-layer feasibility condition is $a_z \le 2n\alpha_z$.

Every contiguous interval inherits the same global floor. Its entering reserve must bound $X_z = \log_z \mathbb{E}z^r$, rather than $\mathbb{E}r$; the universal value $2n$ is safe. A bound on the old mean-rank reserve M_I alone cannot be substituted, because $\log_z \mathbb{E}z^r$ can exceed $\mathbb{E}r$. A smaller proved moment reserve may be used with that interval's own charged gates and noise rounds.

11.7 A stronger consequence at a fixed positive score

For fixed $p > 0$, both b_0 and H_k decay exponentially. Choose $\ln z = 1/\sqrt{k}$ for sufficiently large k, which respects $z \le 2^{1/7}$. The direct polygon then gives

$$a_z \ge 2k - O_p(\sqrt{k}), \qquad \alpha_z = q + O_q(k^{-1/2}).$$

The first estimate follows from $L_z(p) \ge (p - o(1))z^{2k}$; the second follows by expanding the logarithm in (11.7). The recovery bridge also reaches $a_z = 2k - o(k)$ because fixed positive scores give $F_0 \ge k^{-O_p(1)}$. Suppose $q \in (0,1)$ is fixed, $n \ge k$, $D \to \infty$, and $\ln(n/k) = o(D)$. The coasting time in the discrete injection ledger is then $o(D)$, so every feasible family with score bounded away from zero obeys

$$m_T \ge (2 - o(1))\, k\, \frac{1-q}{q}\, D. \tag{11.15}$$

The earlier chord floor gives the corresponding leading coefficient $2p$. The order remains $k\gamma D$ at weak noise, but the necessary rate approaches the full $2k$ payload coefficient at every fixed positive score. The next subsection quantifies vanishing scores; the separate Tanner construction matches resource order in its stated regime, without identifying a numerical optimum.

11.8 An explicit decay guarantee below a necessary rate

For $1 < z \le 2^{1/7}$ the direct branch profile obeys $p \le z^{-2k}\mathbb{E}z^r$, giving the all-cut floor $a_z = [2k + \log_z p]_+$. Set $M = 2n$ and $\alpha = \alpha_z$. For $D \ge 1$, inversion of the discrete injection inequalities gives

$$\begin{aligned} U_z &= \min\left\{ M\alpha,\ \min_{1\le l\le D} \frac{\alpha(m_T + M\alpha^{l-1})}{1 + (D-l)(1-\alpha)} \right\}, \\ p &\le \min\{1, z^{U_z - 2k}\}. \end{aligned} \tag{11.16}$$

Indeed the ledger requires $m_T \ge a_z/\alpha + (D-l)(1-\alpha)a_z/\alpha - M\alpha^{l-1}$ for every l. Rearrangement yields the inner minimum. The width cap after any noise layer gives $a_z \le M\alpha$. If the floor is nonpositive the score inequality already holds. For $D = 0$ use $U_z = M$, giving the trivial bound one. This finite formula retains the full initial width allowance.

Fix $q \in (0,1)$ and finite $\rho \ge 0$. Along any family with $k, D \to \infty$, $n \ge k$, $\ln(n/k) = o(D)$ and $\limsup m_T/(kD) \le \rho$, one obtains

$$\liminf \frac{-\ln p}{k} \ge E_q(\rho), \qquad E_q(\rho) = \sup_{1<z\le 2^{1/7}} (\ln z) \left[2 - \rho \frac{\alpha_z}{1-\alpha_z} \right]_+ . \tag{11.17}$$

For fixed z, choose $l = 1 + \lceil \ln(2n/k)/(-\ln \alpha_z) \rceil = o(D)$, so the surviving initial allowance is at most k. Dividing (11.16) by k proves the fixed-z bound; the same liminf exceeds every such bound, hence their supremum. Use $-\ln 0 = +\infty$.

Because $\alpha_z > q$ and $\alpha_z \to q$ as $z \downarrow 1$, the exponent is positive exactly when $0 \le \rho < 2(1-q)/q$. The left endpoint has continuous value zero; the right endpoint is admitted. Rational-interval

optimization is globally certified by strict concavity: with $t = 2\ln z$, $f(t) = \ln(1 - q + qe^t)$ and $g = t - f > 0$, the untruncated objective is $(1 + \rho/2)t - (\rho/2)t^2/g$, and

$$\left(t^2/g\right)'' = 2(g - tg')^2/g^3 - t^2 g''/g^2 > 0.$$

For $q = 0.99$, certified exponent intervals include $E_q(0.01) \in [0.089980725766, 0.089980725767]$ and $E_q(0.02) \in [0.000050419501, 0.000050419502]$.

Any established nondecreasing concave branch majorant P_z also gives $p \le P_z(z^{U_z})$. The full polygon retains information lost by the simple floor. At $k = 12, n = 120, q = 0.99, z = 27/25, D = 2000, m_T = 200$, it proves $p \le 0.310300082242$, whereas the simple floor gives $p \le 0.418535080865$. This quantitative converse is not an achievable processing capacity; integer refinements or other recovery bounds can remain stronger in particular regimes.

11.9 Propagating the full generating-function profile

The fixed-moment result above replaces every noise step by contraction at one fixed argument. Retaining the argument's exact evolution gives a stronger decay law from the same envelope and guessing bounds.

Let $F(\lambda) = \ln \mathbb{E}e^{\lambda r}$ and define

$$\phi_j(\lambda) = \frac{1}{2}\ln\left[1 + q^j(e^{2\lambda} - 1)\right], \qquad S_q(\lambda) = \sum_{j=1}^{\infty} \phi_j(\lambda).$$

The pivot inequality says a noise layer replaces the profile argument by $\phi_1(\lambda)$, and $\phi_i \circ \phi_j = \phi_{i+j}$. A prescribed T slot adds at most λ to $F(\lambda)$. After subsequent noise layers, its contribution at a cut is bounded by the corresponding $\phi_j(\lambda)$. This retains more information than repeated scalar Jensen contraction. Since $\phi_j(\lambda) = O(q^j)$ on the compact interval used below, the series converges.

The direct branchwise score profile obeys, for $0 \le \lambda \le \ln 2$,

$$p \le e^{-kA(\lambda)}\mathbb{E}e^{\lambda r}, \qquad A(\lambda) = \min\left\{2\lambda, \frac{\ln 2 + 5\lambda}{6}, \ln 2\right\}.$$

The three affine pieces follow from the same output-dimension and quantum-memory bounds; no new uniform guessing exponent is presumed. For $1 \le L \le D$, put $T_L = D - L + 1$. Sum over those noise cuts and charge each prescribed T slot by all of its possible remaining-noise contributions. The finite consequence is

$$\frac{-\ln p}{k} \ge \left[A(\lambda) - \frac{2n\sum_{d=L}^{D}\phi_d(\lambda) + m_T\sum_{j=1}^{T_L}\phi_j(\lambda)}{kT_L}\right]_+.$$

The first term retains the initial full-output width. The second counts fixed circuit slots, rather than an expected executed cost.

Under the same fixed-q growing-family hypotheses as (11.17), the result becomes

$$\boxed{\liminf \frac{-\ln p}{k} \ge \mathcal{E}_q(\rho), \qquad \mathcal{E}_q(\rho) = \max_{0\le\lambda\le\ln 2}[A(\lambda) - \rho S_q(\lambda)]_+.} \tag{11.18}$$

For each fixed argument, take $L = o(D)$ large enough to exhaust the width reserve; $\ln(n/k) = o(D)$ permits this. The truncated series approaches S_q, and $D/T_L \to 1$. Taking the maximum after the fixed-argument limit proves the stated quantifiers.

This exponent is positive exactly for $\rho < 2(1-q)/q$, and $\mathcal{E}_q(0) = \ln 2$. It strictly improves E_q throughout its positive region. On the old interval $0 < \lambda \le \ln 2/7$, the score exponent is still $A = 2\lambda$, while

$$S_q(\lambda) < \lambda \frac{\alpha_{e^\lambda}}{1 - \alpha_{e^\lambda}}.$$

The larger interval also improves the zero-rate endpoint from $2\ln 2/7$ to $\ln 2$. For positive ρ, strict convexity of S_q gives a unique optimum of the untruncated piecewise strictly concave objective, possibly at the kink $\lambda = \ln 2/7$ or endpoint $\ln 2$. Series-tail enclosures and interval optimization provide reproducible global certificates. This strengthens a converse exponent without establishing an achievable processing capacity or changing Lattice's uniform 1/6 exponent.

Finite integer algebra parameters can sharpen the convenient analytic polygon. For $k = 12$, combine the direct quantum-dimension and output-dimension bounds with the established purely classical bound β^{12}, where $\beta = (1 + 1/\sqrt{2})/2$. A rational certificate checks every one of the 169 allowed integer pairs (s, c) with $0 \le 2s + c \le 24$. At $z = 27/25$, the concave polygon through

$$(1, 1/4096), \qquad (z^{11}, 1/4), \qquad (z^{24}, 1),$$

continued constantly afterward, lies above each combined branch cap. The classical cap itself follows by bounding the absolute Gram-matrix row sums of selected BB84 vectors, and is attained by independent Breidbart measurements when enough classical bits are retained.

For the same example $k = 12, p = 0.4, n = 120, q = 0.99$, this certified polygon gives

$$M_z \ge \frac{4z^{11} + z^{24}}{5}, \qquad a_z \simeq 14.84082963, \qquad \alpha_z \simeq 0.99072498.$$

Exact logarithm enclosures put the unrounded individual-gate expression at $243.117317956157\ldots$ for $D = 2000$, requiring at least 244 T/T-dagger gates, compared with the earlier mean-rank count 171. For every 700-round contiguous interval with moment reserve at most 240, the expression is $57.347537146205\ldots$, requiring at least 58 gates instead of 39. These are certified necessary counts, without a claim of physical achievability or optimal choice of z. The analytic moment floors, this finite polygon and the earlier mean-rank bounds remain complementary.

At chance $p = b_0$, discarding the challenge and guessing needs zero gates. Full-register replacement gives $p \le b_0 + (1 - b_0)(1 - \gamma^n)^D$, regardless of gates. The score does not verify noise, controller or side-channel premises. It can witness processing to retain classical as well as quantum information. A remote-preparation wrapper needs its own model- and resource-preserving reduction.

A Finite identities, verification, and notation

A.1 The five-qubit correction truth table

The following table uses syndrome order $s_1 s_2 s_3 s_4$ and the logical representatives specified in Section 8. X_j, Y_j, Z_j act on physical qubit j. The two right columns are the values of (8.3). Listing all sixteen correctable cases fixes the correction rule without a free classical lookup during storage.

Syndrome	Error	f_X	f_Z
0000	I	0	0
1000	X_2	1	0
0100	Z_5	0	1
1100	X_3	1	0
0010	Z_3	0	1
1010	Z_1	0	1
0110	X_4	1	0
1110	Y_3	1	1
0001	X_1	1	0
1001	Z_4	0	1
0101	Z_2	0	1
1101	Y_2	1	1
0011	X_5	1	0
1011	Y_1	1	1
0111	Y_5	1	1
1111	Y_4	1	1

A.2 What finite checks establish

Deterministic finite calculations check the identities most exposed to sign, support, and scheduling mistakes. They enumerate all 67 binary Pauli subspace algebras on two qubits, test channel-rank and support-growth identities, and compare 1,944 exact-rational scalar optimization certificates. Separate checks cover 4,096 three-increment correctable Pauli histories, 12,315 small-tree noise patterns, first-root noise, and simultaneous parent/child workspace allocation.

The five-qubit calculation evaluates every one of the $4^5 = 1024$ Pauli errors, checks the correction polynomials, and finds the numbers of errors leaving a nonidentity logical residual at physical weights zero through five:

$$[0,\ 0,\ 90,\ 210,\ 270,\ 198]. \tag{A.1}$$

In particular, all weight-zero and weight-one errors are corrected. Integer batching and concatenation parameters are checked in 240 finite cases. A separate score-transfer suite checks finite quantum-plus-classical rank cases, unconditional branch mixtures, exact rational recovery-comparison certificates, and independently solved scalar injection programs. Independent moment checks cover all binary subspaces at widths one through three, 34,764 rational moment inequalities, 22,878 pivot and reset checks, and 1,020 T-axis growth checks; exact logarithm enclosures certify the displayed finite gate comparison. These calculations test the stated finite implications; the written factorization and averaging arguments establish the all-cut conclusion.

Additional checks exhaust all two-input/two-output local lookup maps, compare a nonlinear coherent correction channel's full Choi state with its projected-channel formula, and verify invariant, batching, length-ratio and workspace inequalities. They also check 864 exact scalar inversions, all 169 integer algebra pairs in the displayed decay comparison, and rational-interval finite and asymptotic optimization certificates. The extensions also check fixed-level inner-code composition,

profile semigroup and fixed-slot convolution identities, convergent-series tails and the stronger exponent optimization. The small scalar Tanner fixture is not a concrete admissible code instance or a practical threshold certificate.

These checks support the finite algebra and circuits. They do not prove an asymptotic statement by extrapolation. General forgetting follows from the envelope induction; the processing converse follows from channel-rank and scalar inequalities; archives, concatenation and the Tanner construction follow from their stated inherited theorems and induction arguments. No finite test establishes optimality of 42 gates, an optimal hardware threshold, or attainment of the scalar relaxation by a memory circuit.

A.3 Boundary cases

If $D = 0$, an identity memory on k qubits suffices. If $\gamma = 0$, it suffices for any finite duration; the positive-noise batching formulas need not be used. If $k = 0$, the task is vacuous. For finite width, positive noise and at least one noise layer, exact zero infidelity is impossible because complete-register replacement has positive probability. The constructive formulas therefore state nonzero error tolerance. The archive case $J = 1$ uses no archived record, and the concatenated construction $L = 0$ uses no recovery node.

A.4 Notation guide

Symbol	Meaning
n	Total noisy width; called N in the archive construction.
k, $K = 2^k$	Logical qubit count and logical Hilbert-space dimension.
D, γ	Number of full noise layers and complete-depolarization strength.
q, κ	$1 - \gamma$ and $-\ln q$.
m_T	Number of internal physical T and T-dagger gates.
V, r	Binary Pauli envelope and its binary dimension.
$\mathcal{A}_V$	Complex Pauli algebra, of dimension 2^r.
M	Terminal envelope dimension, at most $2n$.
M_I	Upper bound on expected decoded-tail envelope rank at an interval's later cut.
R_t, a, a_p	Expected envelope rank and its fidelity- or guessing-score-derived floor.
δ, b	Extinction-probability allowance $\epsilon/(1 - 4^{-k})$ and survival lower bound $1 - \delta$.
X_z, a_z	Logarithmic exponential rank moment and an all-cut lower bound.
α_z	Noise contraction parameter for the moment potential.
B, b_{rep}	Total structural rank-growth charge and number of archive root replicas, respectively.
τ, h, J	Batch length and number of batches; constants differ among constructions.
n_0, s	Archive data-block width and number of its stabilizer generators.
H_{arc}	Number $s(J - 1)$ of intermediate syndrome bits to archive.
L	Concatenation level of the five-qubit code.